

1. Th1 / IFN-gamma

WHAT YOU SEE

Soldier labeled Th1 and IFN-gamma fountain

WHAT IT ENCODES

Sarcoid granuloma formation is a CD4 Th1-driven process. After an antigen is presented on MHC II, CD4 cells secrete IL-2 (T-cell proliferation) and IFN-gamma, which activates macrophages and drives them to fuse into epithelioid and multinucleated giant cells. TNF-alpha from macrophages maintains the granuloma — this is why anti-TNF agents (infliximab) work in refractory disease.

BOARD TRAP

Calling sarcoid a Th2/eosinophil-driven or antibody-mediated disease — it is a Th1/IFN-gamma, cell-mediated granulomatous process. The exaggerated cellular immune response in the tissue paradoxically coexists with peripheral cutaneous ANERGY (negative PPD/skin tests), a classic board point.

2. Non-caseating granuloma

WHAT YOU SEE

Large flower with CD4 petals, tight green checkmark center

WHAT IT ENCODES

The histologic hallmark is the NON-caseating (no central necrosis) granuloma — a tight, well-formed collection of epithelioid macrophages and giant cells. Inclusions you may see (non-specific): asteroid bodies (star-shaped) and Schaumann bodies (laminated calcium). Diagnosis is one of EXCLUSION: you must rule out TB and fungal infection (stains/cultures) before calling non-caseating granulomas sarcoid.

BOARD TRAP

Diagnosing sarcoid on granuloma histology ALONE without excluding infection — non-caseating granulomas are not specific (also seen in berylliosis, Crohn, hypersensitivity pneumonitis, fungal). The board answer: confirm with biopsy AND exclude TB/fungal with AFB/GMS stains and cultures before treating with immunosuppression.

3. CD4 core

WHAT YOU SEE

CD4 labels on the inner petals

WHAT IT ENCODES

The granuloma center is packed with CD4 helper T cells, epithelioid macrophages, and multinucleated giant cells. This local CD4 accumulation in tissue is mirrored by a high CD4:CD8 ratio (>3.5) on bronchoalveolar lavage, a supportive diagnostic finding.

BOARD TRAP

Confusing the tissue CD4 excess with HIV — in sarcoid, CD4 cells are sequestered in granulomas, often causing relative peripheral lymphopenia and cutaneous anergy, but this is NOT HIV. The high CD4:CD8 ratio is in the LUNG (BAL), not the blood.

4. CD8 outer rim

WHAT YOU SEE

CD8 labels on the outer ring

WHAT IT ENCODES

CD8 cytotoxic T cells and fibroblasts form the peripheral rim of the granuloma. Over time, ongoing fibroblast activity and collagen deposition at this rim can lead to the irreversible pulmonary fibrosis seen in end-stage (Stage IV) disease.

BOARD TRAP

Assuming all granulomatous inflammation is reversible — once fibrosis (Stage IV) develops at the granuloma periphery, steroids will not reverse it. The therapeutic window is the inflammatory phase; established honeycomb fibrosis is treated supportively (transplant for end-stage).

5. Caseating = NOT sarcoid

WHAT YOU SEE

Dark necrotic flower with red X labeled necrosis

WHAT IT ENCODES

A CASEATING (central necrosis) granuloma points AWAY from sarcoid and toward infection — classically tuberculosis, also endemic fungi (histoplasma, coccidioides, blastomyces). The necrotic center on biopsy mandates AFB and fungal stains/cultures before any diagnosis.

BOARD TRAP

Diagnosing sarcoid on a caseating granuloma — caseation = think TB/fungal, not sarcoid. The dangerous error is starting steroids/anti-TNF (immunosuppression) for presumed sarcoid in a patient who actually has TB, which would disseminate the infection. Always exclude TB first.

6. Loose granuloma = HP

WHAT YOU SEE

Pale loose flower labeled HP/loose

WHAT IT ENCODES

Loose, poorly-formed, non-necrotizing granulomas centered on bronchioles suggest hypersensitivity pneumonitis (HP) rather than the tight, well-formed granulomas of sarcoid. HP history involves antigen exposure (birds, mold/farmer's lung) and improves on antigen avoidance.

BOARD TRAP

Calling poorly-formed peribronchiolar granulomas with an exposure history 'sarcoid' — that pattern plus a relevant antigen exposure is hypersensitivity pneumonitis. Also note HP shows a LYMPHOCYTE-predominant BAL with LOW CD4:CD8 (the opposite of sarcoid's high ratio).

7. Bilateral hilar adenopathy

WHAT YOU SEE

Lungs at top, >90% with 50% extra label

WHAT IT ENCODES

Bilateral hilar lymphadenopathy is the most common radiographic finding and defines Stage I disease; the thorax is involved in >90% of patients. Extrapulmonary involvement is present in roughly half. The symmetric, bilateral nature of the adenopathy is a key discriminator.

BOARD TRAP

Reading bilateral hilar adenopathy as malignancy/lymphoma or assuming it needs treatment — symmetric BILATERAL hilar adenopathy strongly favors sarcoid (lymphoma/carcinoma tends to be unilateral/asymmetric or mediastinal). Asymptomatic Stage I adenopathy is OBSERVED, not treated.

8. Mediastinum CD4:CD8 >3.5

WHAT YOU SEE

Ninja gauge labeled CD4:CD8 >3.5, EBUS

WHAT IT ENCODES

A bronchoalveolar lavage CD4:CD8 ratio >3.5 is fairly specific for sarcoid and supports the diagnosis. EBUS-TBNA (endobronchial ultrasound–guided transbronchial needle aspiration) of mediastinal/hilar nodes is the preferred, high-yield, minimally invasive way to get tissue.

BOARD TRAP

Going straight to surgical/mediastinoscopy or open lung biopsy first — EBUS-TBNA of accessible nodes is the lower-risk first-line tissue diagnostic. Also, a CD4:CD8 ratio that is LOW points elsewhere (HP, drug reaction), not sarcoid.

9. MHC II

WHAT YOU SEE

Purple antigen-presenting cell labeled MHC II

WHAT IT ENCODES

An (unknown) antigen is processed and presented on MHC class II by antigen-presenting cells (macrophages/dendritic cells) to CD4 T cells, initiating the Th1 cascade and granuloma formation. Specific HLA alleles modify risk and phenotype: HLA-DRB1*0301 → Löfgren/good prognosis; HLA-DRB1*1501 → chronic/worse disease.

BOARD TRAP

Thinking sarcoid has a single identified causative antigen — the inciting antigen remains UNKNOWN. It is a multifactorial gene–environment disease; the HLA-DRB1 associations are about susceptibility and prognosis, not a defined trigger.

10. African-American severe

WHAT YOU SEE

A Black woman in the foreground tagged '35.5' (more severe) — she IS the highest-yield sarcoid vignette: the African-American woman with the worst, chronic multi-organ prognosis.

WHAT IT ENCODES

In the US, sarcoidosis is most prevalent in African-Americans (especially women), who tend to have more severe, chronic, multi-organ disease (cardiac, ocular, lupus pernio, chronic skin) and higher mortality. Classic boards vignette: a young African-American woman with cough, dyspnea, and bilateral hilar adenopathy.

BOARD TRAP

Forgetting the demographic skew or assuming uniform prognosis across groups — the highest-yield exam stem is the African-American woman, and African-American race itself is a poor-prognosis marker. Don't undertreat severe multi-organ disease in this population by anchoring on the benign Löfgren outcome.

11. Scandinavian / age peaks

WHAT YOU SEE

Two pale Scandinavian-looking figures tagged '24' — the young age tag = the first (25-35) of sarcoid's bimodal age peaks.

WHAT IT ENCODES

Sarcoid has a bimodal age distribution: a first peak around 25-35 years and a second peak in women over 50. Northern European/Scandinavian populations are also notably affected and more often present with the acute, good-prognosis Löfgren phenotype.

BOARD TRAP

Restricting sarcoid to young adults only — the second peak in older women (>50) is real, so don't exclude sarcoid in an older woman with hilar adenopathy and assume malignancy. Conversely, the Scandinavian Löfgren presentation tends to remit, unlike the chronic African-American phenotype.

12. Spring magnifier

WHAT YOU SEE

A wooden sign reading SPRING beside a magnifying glass — spring season under the lens = the seasonal/environmental clustering pointing to an inhaled trigger.

WHAT IT ENCODES

Sarcoid incidence shows seasonal clustering (spring/early summer) and geographic clustering, supporting an environmental/infectious antigen trigger acting on a genetically susceptible host. Proposed (unproven) triggers include mycobacterial/propionibacterium antigens and inorganic dusts.

BOARD TRAP

Treating seasonal clustering as proof of a specific infectious cause or as a reason to give antibiotics — the trigger is unidentified and sarcoid is not treated as an infection. Granulomatous diseases from defined organisms (TB, fungal) must still be excluded by stain/culture.

13. 9/11 WTC exposure

WHAT YOU SEE

Firefighter labeled 9/11 WTC 86

WHAT IT ENCODES

Occupational/environmental inhalational exposures are linked to sarcoid-like granulomatous disease — World Trade Center responders showed increased rates of pulmonary granulomatous disease. The strongest occupational mimic is chronic beryllium disease (berylliosis), which is histologically identical to sarcoid.

BOARD TRAP

Calling a beryllium-exposed worker's lung granulomas 'sarcoid' — chronic beryllium disease looks identical on biopsy. The discriminator is occupational history (aerospace, electronics, metal alloys) plus a POSITIVE beryllium lymphocyte proliferation test (BeLPT). Missing the exposure history and labeling it sarcoid is the trap.

14. Baseline imaging

WHAT YOU SEE

Shrine with chest films labeled BASELINE

WHAT IT ENCODES

At diagnosis, obtain a baseline workup for staging and to screen organ involvement: chest X-ray/CT (staging), PFTs with DLCO, EKG (cardiac screen), ophthalmologic slit-lamp exam, serum calcium/24h urine calcium, renal and liver function, and CBC. ECG/echo ± cardiac MRI if cardiac symptoms; serum ACE is supportive but neither sensitive nor specific.

BOARD TRAP

Relying on an elevated serum ACE to diagnose or monitor sarcoid — ACE is neither sensitive nor specific (false positives in DM, hyperthyroidism, other granulomatous disease; normal in many sarcoid patients). It is NOT a diagnostic or reliable monitoring test on its own; tissue biopsy plus exclusion of mimics is the diagnostic standard.